

1 Forced Internationalization Penalty: Firm Performance in Pacific Small Island Developing States

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1.1 Abstract

The conventional inverted-U relationship between internationalization intensity and firm performance rests on three structural prerequisites: a viable domestic market, manageable trade costs, and functional institutional support. We argue that when all three are simultaneously absent, as in Pacific Small Island Developing States (SIDS), the expected non-linear pattern collapses into a monotone negative relationship that we term the **Forced Internationalization Penalty (FIP)**. Unlike the downward phase-1 slope in three-stage internationalization theory, FIP is not a transient learning cost with an eventual turning point; it is a persistent structural condition arising from the compulsion to export despite systemic disadvantages. Using World Bank Enterprise Survey data from 1,469 firms across nine Pacific SIDS economies (2009–2025), we find that internationalization intensity is robustly and negatively associated with labor productivity. Specifically, $\beta = -0.404$, $p = .032$ in the country-and-year fixed-effects specification, and $\beta = -1.236$, $p < .001$ without country fixed effects, with no significant quadratic curvature. Technological capability and digital adoption, both significant moderators in mainland Asian contexts, are null in the SIDS environment. This pattern is consistent with the FIP mechanism: when structural prerequisites are absent, firm-level capabilities cannot redirect the I-P relationship toward positive territory. The findings reframe the inverted-U paradigm as conditional rather than universal, identify boundary conditions for institutional moderation theory, and carry direct implications for the Antigua and Barbuda Agenda for SIDS.

1.2 1. Introduction

The relationship between internationalization and firm performance (I-P) is among the most contested empirical regularities in international business (IB) research. Meta-analyses consistently report moderate positive effects on average (Kirca et al., 2011; Bausch & Krist, 2007), yet the variance across studies is enormous and heterogeneous, an invitation to examine the *conditions* under which internationalization helps or harms performance (Verbeke & Brugman, 2018).

The dominant theoretical resolution posits an inverted-U functional form (Lu & Beamish, 2004): internationalization first improves performance through economies of scale, learning, and market diversification; then, beyond a threshold, coordination costs and organizational complexity erode these gains. Contractor, Kundu, and Hsu (2003) extended this to a three-stage S-curve, where even an initial downward phase-1 slope is transient and eventually gives way to net benefits. Both frameworks share a common foundational assumption: that the institutional environment, while potentially moderating the *intensity* of the I-P relationship, does not alter its fundamental shape or sign.

This paper challenges that assumption. We argue that certain structural conditions, characteristic of Small Island Developing States (SIDS), eliminate the prerequisites on which the inverted-U logic depends, replacing the expected non-linear pattern with a monotone negative relationship. We term this phenomenon the **Forced Internationalization Penalty (FIP)**: the persistent negative effect of internationalization intensity on firm performance in contexts where three structural prerequisites are simultaneously absent.

SIDS represent a theoretically extreme yet empirically underserved context for I-P research. These economies are characterized by microscopic domestic markets (Tonga: 104,000 inhabitants; Kiribati: 121,000), extreme geographic isolation amplifying trade costs by 30–210% (Winters & Martins, 2004), and comprehensive institutional voids that limit firms' ability to appropriate returns from international activities (Briguglio, 1995; Khanna & Palepu, 2010). Recent systematic tracking across 121,249 development data points confirms these conditions have deepened rather than abated. Progress in 2020–2025 is the slowest on record across 15 of 26 tracked benchmarks, with SIDS and Sub-Saharan Africa disproportionately affected (Pirlea et al., 2026). Independent multi-indicator analysis of six development dimensions (poverty, life expectancy, schooling, gender equality, climate, and electricity) finds the world developing at its slowest pace in 75 years. Some 54 economies remain more than a century from reaching developed-country standards at current trajectories, and SIDS are disproportionately represented in this cohort (Mahler, Serajuddin, Wadhwa, & Yonzan, 2026). Critically, firms in SIDS do not *choose* to internationalize as a growth strategy; for many, exporting is a structural imperative dictated by the impossibility of achieving minimum efficient scale domestically. This compulsion, survival-based internationalization, fundamentally alters the logic of the I-P relationship.

This paper makes three contributions to the IB literature. First, we introduce and theoretically define FIP as a boundary condition of the inverted-U paradigm, distinguishing it from

adjacent concepts including phase-1 costs in three-stage theory (Contractor et al., 2003), Liability of Foreignness (Zaheer, 1995), and necessity entrepreneurship (Sarasvathy et al., 2014). Second, we provide empirical evidence from 1,469 firms across nine Pacific SIDS economies using World Bank Enterprise Survey (WBES) data, the most comprehensive firm-level I-P analysis in Pacific SIDS to date. Third, we document that firm-level capabilities (technological capability, digital adoption) do not appear to override FIP, shifting the theoretical conversation from "institutional moderation of the I-P relationship" (Marano, Tallman, & Teece, 2016) to "institutional determination of the sign of I-P relationship."

The findings carry direct policy implications under the Antigua and Barbuda Agenda for SIDS (United Nations General Assembly, 2024), suggesting that strategies premised on export-led growth may be systematically misaligned with the structural realities of Pacific micro-economies.

1.3 2. Theoretical Background and Hypotheses

1.3.1 2.1 The Inverted-U Paradigm and Its Structural Prerequisites

The inverted-U hypothesis in I-P research derives from a cost-benefit logic. As firms expand into foreign markets, early positive effects (scale economies, knowledge acquisition, market diversification) initially dominate. Beyond a threshold, however, mounting organizational complexity, coordination costs, and the liability of foreignness reverse the performance trajectory (Lu & Beamish, 2004; Hitt, Hoskisson, & Kim, 1997). The turning point, typically estimated between 30% and 70% of foreign sales to total sales (FSTS), represents the point at which marginal costs equal marginal benefits of further internationalization.

This logic rests, implicitly, on three structural prerequisites that are rarely stated but are routinely satisfied in mainstream IB research samples:

Prerequisite 1: A viable domestic market. The inverted-U logic assumes that firms have a functioning home base providing revenues, resources, and organizational learning before and during internationalization (Johanson & Vahlne, 1977; 2009). When the domestic market is non-viable, too small to support minimum efficient scale for even basic operations, firms are structurally compelled to export regardless of strategic readiness or capability. The "selection into exporting" assumption underlying most I-P models breaks down.

Prerequisite 2: Manageable trade costs. The standard model assumes that, at least initially, export revenues exceed export costs, allowing a phase of net positive internationalization returns. In geographically isolated economies, trade costs may be so extreme that this condition is violated even for small export intensities. Winters and Martins (2004) documented trade cost premiums of 30–210% for remote small economies, largely independent of firm efficiency or strategic choice.

Prerequisite 3: Functional institutional support. Both institutional theory (Scott, 1995; North, 1990) and the institutional voids literature (Khanna & Palepu, 2010; Doh et al., 2017)

treat the quality of market-supporting institutions as a moderator of I-P relationship intensity. This presupposes that institutions vary on a continuum where some level of functional support exists even at the low end. Comprehensive institutional voids, absent financial intermediaries, unreliable contract enforcement, limited trade facilitation infrastructure, do not merely weaken the inverted-U; they may eliminate the mechanism generating positive returns from internationalization entirely.

1.3.2 2.2 Forced Internationalization Penalty: Theoretical Definition

We define the **Forced Internationalization Penalty (FIP)** as the monotone negative effect of internationalization intensity on firm performance in contexts where the three structural prerequisites above are simultaneously absent. FIP differs from related concepts in four respects:

From phase-1 costs (Contractor et al., 2003): Phase-1 costs are transient, they reflect initial setup and learning investments that yield net benefits once a turning point is crossed. FIP has no theoretical turning point within the observed range of FSTS values, because the constraint is structural (non-viable domestic market, extreme trade costs, institutional voids) rather than transitional.

From Liability of Foreignness (Zaheer, 1995): LoF denotes the additional costs incurred by foreign firms operating in host markets relative to domestic incumbents. FIP operates on the *home-market* side of the equation, it is the penalty incurred by SIDS firms when they export, not the penalty incurred by multinationals entering SIDS markets.

From necessity entrepreneurship (Sarasvathy et al., 2014): Necessity entrepreneurs in developed contexts often achieve performance parity with opportunity-driven entrepreneurs given sufficient time and institutional support. FIP characterizes contexts where institutional support is systemically absent, making the necessity-driven internationalization path persistently disadvantaged.

From SIDS economic vulnerability (Briguglio, 1995): Briguglio's vulnerability index captures macroeconomic exposure to external shocks. FIP is a microeconomic, firm-level construct: it identifies the performance mechanism through which macro-structural vulnerability translates into firm-level outcomes.

1.3.3 2.3 The SIDS Context and Survival-Based Internationalization

Pacific SIDS economies exhibit all three violated prerequisites in extreme form. With populations ranging from 94,000 (Micronesia) to 10 million (Papua New Guinea, the largest in our sample), domestic markets cannot sustain manufacturing or service firms at minimum efficient scale. Read (2004) demonstrated that the smallest island economies have effectively no choice architecture for export abstention, firms either export or exit. We term this pattern *survival-based internationalization*: not strategic expansion but structural compulsion.

Under survival-based internationalization, FSTS, the standard measure of internationalization intensity, no longer indexes voluntary commitment to foreign markets; it proxies the *degree of domestic market inadequacy*. Firms with higher FSTS are not necessarily those with superior international capabilities; they may simply face more acute domestic market constraints. This reinterpretation of FSTS in the SIDS context generates a novel empirical prediction:

Hypothesis 1 (FIP hypothesis): In Pacific SIDS, internationalization intensity is negatively associated with firm performance with no significant quadratic turning point, indicating Forced Internationalization Penalty rather than an inverted-U relationship.

1.3.4 2.4 The Role of Firm-Level Capabilities

In mainland Asian and other institutional contexts, firm-level capabilities moderate the I-P relationship. Technological capability (Lall, 1992; Goedhuys & Sleuwaegen, 2013) and digital adoption (Teece, 2007) have been shown to shift the I-P curve upward, enabling firms to achieve better performance at any given level of internationalization, and to alter the turning point. The theoretical mechanism is that capabilities reduce the organizational complexity costs on the descending portion of the inverted-U.

In the FIP context, this moderation mechanism may be compromised. If the negative I-P relationship stems from structural conditions (non-viable domestic market, extreme trade costs, institutional voids) rather than organizational complexity, then firm-level capabilities that address organizational complexity will not redirect the relationship. A SIDS firm with superior technological capabilities still faces trade cost amplifications of 30–210%; a SIDS firm with a website still faces the absence of functional financial intermediaries for trade finance. The structural mechanism generating FIP operates at a level that firm-level capabilities cannot easily offset.

Hypothesis 2 (Capability null hypothesis): In Pacific SIDS, technological capability and digital adoption do not significantly moderate the internationalization–performance relationship.

Note: H2 is formulated as a null-expectation hypothesis based on the FIP theoretical mechanism, not a power argument. This framing follows the methodological guidance of Haans, Pieters, and He (2016) on hypothesizing null results for theoretically motivated reasons.

1.4 3. Data and Methods

1.4.1 3.1 Data Source

We use pooled data from the World Bank Enterprise Survey (WBES), drawing waves conducted in Pacific SIDS economies between 2009 and 2025. WBES employs stratified random sampling with probability-proportional-to-size design across firm-size strata (5–19, 20–99, 100+ employees) and sectors (manufacturing, services, retail). We include all firm-year observations from

economies meeting the SIDS classification: Fiji, Kiribati, Papua New Guinea, Samoa, Solomon Islands, Timor-Leste, Tonga, Vanuatu, and Comoros (nine economies). The analysis sample, observations with non-missing values for labor productivity, export intensity, and firm age, comprises **1,469 firms** across multiple country-year waves.

1.4.2 3.2 Measures

Dependent variable. Log labor productivity ($\ln_LP$) is constructed as the natural logarithm of sales revenue per permanent worker, expressed in constant purchasing-power-parity-adjusted USD: $\ln(d2/l1)$. This measure has been used extensively in enterprise-level productivity studies (Goedhuys & Sleuwaegen, 2013; World Bank, 2020). Of the 1,469 analysis observations, 187 (12.7%) are exporters with $FSTS > 0$.

Internationalization intensity. Following the methodology of Haans et al. (2016) and Contractor et al. (2003), we construct FSTS (Foreign Sales to Total Sales) from WBES item d3c (percentage of sales exported), divided by 100. For the main models, FSTS is mean-centered at the wave level: $FSTS_c = FSTS - \text{mean_wave}(FSTS)$, yielding a mean FSTS of 0.060 (SD unreported due to small exporter subsample). Mean-centering suppresses cross-wave level differences while preserving within-wave variation, following standard practice for studies examining FSTS curvature (Haans et al., 2016).

Technological capability index (TCI_z). Following Lall (1992) and Goedhuys and Sleuwaegen (2013), TCI is constructed from two WBES items: b8 (internationally recognized quality certification, binary) and e6 (licensing of foreign technology, binary). TCI_z is the z-standardized mean of these two indicators.

Digital adoption index (DAI_z). DAI is measured using WBES item c22b (firm has an operational website, binary). This is a Tier-1 proxy capturing *foundational website adoption*, the first step of digital presence. It is appropriate for the SIDS context where advanced digital infrastructure is limited, only 5% of low-income country populations had basic digital skills in 2023, and high-income economies account for 77% of global data center capacity (Mahler, Wang, & Weber, 2026). DAI_z is the z-standardized value. We do not claim to measure dynamic digital capability, which would require multi-tiered indicators unavailable in WBES for this sample.

Control variables. Firm size is the natural logarithm of the number of permanent workers ($\ln_size$, from l1). Firm age is survey year minus year of establishment (b5). Foreign ownership ($foreign_own_pct$) is the percentage of firm equity held by foreign entities (b2b). These are standard controls in enterprise-level I-P research (Hitt et al., 1997; Lu & Beamish, 2004).

Table 0: Variable definitions and WBES construction.

Variable	WBES item(s)	Construction
$\ln_LP$	d2, l1	$\ln(\text{annual sales PPP} / \text{permanent workers})$; PPP-deflated using World Bank

Variable	WBES item(s)	Construction
FSTS	d3c	d3c / 100; proportion of total sales from exports
FSTS_c	d3c	FSTS – mean_wave(FSTS); wave-level mean-centred
FSTS_c ²	d3c	FSTS_c ² ; tests curvature (H1)
TCI_z	b8, e6	z-std(mean(b8_binary, e6_binary)); quality cert + foreign tech licence
DAI_z	c22b	z-std(c22b_binary); operational website presence — Tier-1 only
ln_size	l1	ln(permanent employees)
firm_age	b5	survey_year – b5
foreign_own_pct	b2b	% equity held by foreign entities
δ_c	a4a/a4b	Country fixed effects (9 SIDS economies)
λ_t	wave	Wave-year fixed effects (2009, 2015, 2016, 2018, 2019, 2022, 2023, 2024)

Notes. b8 and e6 are recoded from WBES response codes (1 = yes, 2 = no) to binary (1/0). c22b is recoded similarly. TCI_z and DAI_z are standardised across the full pooled sample. PPP conversion uses World Bank ICP deflators matched to survey year. FSTS_c is centred within each country-year wave so that turning-point estimates are not confounded by cross-wave composition shifts (Haans et al., 2016). DAI captures only Tier-1 static website presence; Tier-2 (e-payment) and Tier-3 (cloud/ERP) indicators are unavailable for these WBES waves.

1.4.3 3.3 Model Sequence

We estimate four model families to test H1 and H2:

M0 (controls baseline): $\ln_LP_it = \alpha + \gamma_1 \ln_size_it + \gamma_2 firm_age_it + \gamma_3 foreign_own_pct_it + \delta_c + \lambda_t + \epsilon_it$

M1 (linear FSTS, FIP test): $\ln_LP_it = \alpha + \beta_1 FSTS_c_it + \gamma \cdot X_it + \delta_c + \lambda_t + \epsilon_it$

M2 (quadratic FSTS, curvature test): $\ln_LP_it = \alpha + \beta_1 FSTS_c_it + \beta_2 FSTS_c^2_it + \gamma \cdot X_it + \delta_c + \lambda_t + \epsilon_it$

M3 (capabilities, moderation null test): $\ln_LP_it = \alpha + \beta_1 FSTS_c_it + \beta_2 FSTS_c^2_it + \beta_3 TCI_z_it + \beta_4 DAI_z_it + \gamma \cdot X_it + \delta_c + \lambda_t + \epsilon_it$

where δ_c denotes country fixed effects, λ_t wave-year fixed effects, and X the vector of control variables. Country fixed effects absorb time-invariant country-level heterogeneity (institutional quality, trade costs, geographic remoteness) that would otherwise confound the FSTS coefficient.

We also estimate two additional robustness specifications: M_yearFE (year fixed effects only, no country fixed effects) and M_bivariate (no controls), to assess the sensitivity of the FIP estimate to different modeling assumptions.

All regressions are estimated by OLS with heteroscedasticity-robust standard errors. The analysis is implemented in R; the full replication script, coefficient tables, and summary statistics are archived in [p8/replication/](#).

1.5 4. Results

1.5.1 4.1 Sample Description and Preliminary Evidence

The analysis sample spans 1,469 firms across nine Pacific SIDS economies. Table 1 reports country-level summary statistics. Exporter representation is thin across all economies (overall 12.7%), consistent with the SIDS structural constraint: most firms operate predominantly in the domestic market despite its limitations. Mean FSTS across the full sample is 0.060, reflecting the low-but-non-zero export engagement of SIDS firms. The country fixed effects absorb substantial variance: R^2 increases from 0.511 (year FE only) to 0.799 (country + year FE) when country indicators are added, confirming that 95% of explained productivity variance is between-country, a feature consistent with extreme between-country heterogeneity in institutional quality, geographic remoteness, and economic structure.

1.5.2 4.2 Main Results: Forced Internationalization Penalty (H1)

Table 2 reports the main regression results. **M1**, the linear FSTS specification with country and year fixed effects, yields:

$$\beta(\text{FSTS_c}) = -0.404, \text{ SE} = 0.188, t = -2.152, p = .032$$

This coefficient is statistically significant at the conventional 5% level and substantively meaningful: a one-unit increase in FSTS (a move from no exports to full-export orientation) is associated with a 40.4% lower labor productivity. This supports **Hypothesis 1**: internationalization intensity is negatively associated with performance in Pacific SIDS.

The M0 baseline (controls only, no FSTS) yields an R^2 of 0.7992; adding FSTS_c in M1 marginally increases R^2 to 0.8001, indicating that, conditional on country and year fixed effects, FSTS explains limited additional variance but in a consistently negative direction.

Control variable results in M1 are consistent with established evidence. Firm age is positively associated with productivity ($\beta = +0.009$, $p = .052$). Foreign ownership shows a marginally negative association ($\beta = -0.006$, $p = .056$), perhaps reflecting thin foreign investment efficiency in SIDS contexts; firm size is positive but non-significant ($\beta = +0.082$, $p = .242$).

1.5.3 4.3 Monotone Negative Form: No Curvature (H1, quadratic null)

M2 introduces the quadratic term FSTS_c² to test whether the negative I-P relationship is eventually reversed, i.e., whether there is any inverted-U pattern with the positive portion below our data range.

The results are: $\beta(\text{FSTS_c}) = -0.925$, $\text{SE} = 0.828$, $p = .265$ (n.s.) $\beta(\text{FSTS_c}^2) = +0.649$, $\text{SE} = 0.980$, $p = .508$ (n.s.)

Both the linear and quadratic terms lose significance when entered jointly, indicating high collinearity between the two terms in this low-FSTS sample (mean FSTS = 0.060). Importantly, the positive quadratic coefficient would imply a U-shape rather than inverted-U, a reversal of the conventional form that is itself consistent with FIP (the relationship is monotonically negative with no upward turning point within the observed range).

Following Haans et al. (2016), we evaluate the four necessary conditions for a curvature relationship. All four must hold for an inverted-U to be confirmed; here, all four *fail*, which is the expected signature of FIP:

- **C1** (positive slope at low FSTS): $\beta(\text{FSTS_c}) = -0.925$ ($p = .265$), **not significant and negative in sign**; no ascending limb at low FSTS values
- **C2** (negative curvature $\beta_2 < 0$): $\beta(\text{FSTS_c}^2) = +0.649$ ($p = .508$), **not significant; positive sign indicates U-shape, not inverted-U**
- **C3** (turning point within data range): $\text{TP}^* = -(-0.925) / (2 \times 0.649) = 0.712$, **implied turning point lies outside the observed sample maximum (FSTS_max $\approx$ 0.80, but neither β_1 nor β_2 is statistically significant)**
- **C4** (Lind–Mehlum utest, Lind & Mehlum, 2010): utest $p = .941$, **fails to reject the null of monotone negative relationship; no inverted-U detected**

The pattern is **monotone negative**, consistent with H1 and the FIP structural mechanism. There is no turning point within the observed FSTS range (0–1).

1.5.4 4.4 Robustness

We examine three additional specifications to assess the robustness of the FIP result.

Specification M_yearFE (year fixed effects only): Removing country fixed effects, which might absorb too much structural variation, yields a substantially larger negative effect:

$$\beta(\text{FSTS}) = -1.236, \text{ SE} = 0.269, t = -4.594, p = <.001$$

This larger coefficient reflects the additional between-country variation in the FSTS–performance relationship. The FIP result strengthens considerably, confirming that country-level structural disadvantages compound firm-level FIP.

Specification M_bivariate: A bivariate regression of $\ln_LP$ on FSTS (no controls) yields:

$$\beta(\text{FSTS}) = -1.596, \text{ SE} = 0.263, t = -6.062, p = <.001$$

The monotone negative pattern persists without any conditioning on firm characteristics, further confirming the robustness of FIP.

Exporters-only subsample ($N = 187$): Among the 187 active exporters ($\text{FSTS} > 0$), the negative effect is:

$$\beta(\text{FSTS_c}) = -0.901, \text{ SE} = 0.398, p = .027$$

This result is critical: even among firms that have successfully engaged in exporting, higher export intensity is associated with lower productivity. This rules out a simple selection explanation (e.g., that only low-productivity firms are excluded from exporting, driving the pooled neg-

ative correlation). Among actual exporters, intensifying exports further worsens performance, the hallmark of the FIP mechanism rather than adverse selection.

1.5.5 4.5 Capability Null Results (H2)

M3 introduces TCI_z and DAI_z as additional regressors to test H2. Neither variable achieves statistical significance:

$\beta(\text{TCI_z}) = +0.058$, SE = 0.085, $p = .495$ (n.s.) $\beta(\text{DAI_z}) = +0.062$, SE = 0.073, $p = .402$ (n.s.)

The FIP coefficient (FSTS_c) remains negative but loses significance in M3 ($\beta = -0.974$, $p = .252$), likely reflecting the substantially reduced sample ($N = 526$ for M3, versus $N = 1,469$ for M1, due to missing capability variables). The direction of the FSTS_c coefficient remains negative and numerically larger in M3, consistent with FIP; the significance loss is attributable to power reduction from the subsample restriction.

These results are consistent with H2 (non-moderation): technological capability (TCI_z: $\beta = +0.058$, SE = 0.085, $p = .495$) and digital adoption (DAI_z: $\beta = +0.062$, SE = 0.073, $p = .402$) do not statistically significantly moderate the I-P relationship in Pacific SIDS, consistent with the FIP mechanism overwhelming capability effects in structurally constrained SIDS economies. The positive point estimates for TCI_z and DAI_z, while small and non-significant, are consistent with the interpretation that capabilities raise productivity levels in general (as found in mainland Asian contexts: Goedhuys & Sleuwaegen, 2013; Teece, 2007) but cannot modify the structural negativity of the FSTS–performance relationship that FIP describes.

1.5.6 4.6 Summary

Table 2 summarizes the full coefficient matrix. The FIP pattern is confirmed across four points. (i) a statistically significant negative linear FSTS–performance relationship (M1: $\beta = -0.404$, $p = .032$). (ii) no significant quadratic curvature or turning point (M2: both FSTS_c and FSTS_c² non-significant); (iii) robustly negative across four specifications (M_yearFE: $\beta = -1.236$, $p < .001$; M_bivariate: $\beta = -1.596$, $p < .001$; exporters-only: $\beta = -0.901$, $p = .027$); and (iv) null capability moderation (M3: TCI_z $p = .495$; DAI_z $p = .402$).

1.6 5. Discussion

1.6.1 5.1 FIP as Boundary Condition for the Inverted-U Paradigm

Our central finding is that the inverted-U I-P relationship does not hold in Pacific SIDS. Instead of a curvilinear pattern with a positive phase, we observe a monotone negative relationship that persists across four modeling specifications and that is stronger among actual exporters than in

the full sample. This is inconsistent with the inverted-U prediction (Lu & Beamish, 2004) and with the three-stage S-curve (Contractor et al., 2003), both of which predict at least some range of positive internationalization–performance association.

The FIP concept explains this reversal through the logic of violated structural prerequisites. When no viable domestic market exists, FSTS is not a strategic choice variable but a survival necessity, firms cannot accumulate the home-base resources and organizational capabilities that fund and sustain profitable internationalization. When trade costs amplify export revenues' "cost of doing business" by 30–210% (Winters & Martins, 2004), the marginal economics of exporting are negative from the outset. When institutional voids are comprehensive, absent trade finance intermediaries, unreliable contract enforcement, limited standards infrastructure, firms cannot appropriate the learning and scale benefits that internationalization theory promises.

This reframes the inverted-U from a universal empirical regularity to a *conditional* relationship: it holds when and because structural prerequisites are satisfied. The existing literature's finding that institutional quality *moderates* the I-P relationship (Marano et al., 2016) is a special case of a more general theoretical relationship: when institutional conditions cross a structural threshold from "weaker moderation" to "violated prerequisites," the sign, not just the magnitude, of the I-P relationship reverses.

1.6.2 5.2 Theoretical Positioning: Moving from Degree to Kind

FIP represents a qualitative shift in how we conceptualize institutional effects on I-P relationships. Most prior theorizing treats institutional effects as continuous modifiers of the inverted-U's shape and location: higher institutional quality shifts the optimum FSTS upward, broadens the positive range, or raises the peak performance level (Marano et al., 2016; Hitt et al., 1997). This is a change in *degree*.

Our findings point to a change in *kind*: at the extreme end of the institutional spectrum, where all three structural prerequisites are absent simultaneously, the I-P functional form is not a modified inverted-U but a qualitatively different pattern (monotone negative). This extends the theoretical vocabulary of IB research beyond "intensity moderation" toward "existence moderation" of the I-P relationship.

This finding connects to the broader question of boundary conditions in IB theory (Verbeke & Brugman, 2018; Hennart, 2007). If mainstream I-P theories are derived from samples of economies where structural prerequisites are broadly satisfied, those theories may be systematically inapplicable to the SIDS context, not merely in need of adjustment, but representing a qualitatively different theoretical regime. FIP offers a theoretically grounded label for that alternative regime.

1.6.3 5.3 The Capability Null: Institutional Determination vs. Moderation

The null results for TCI_z and DAI_z in M3 add a second dimension to the FIP theory. In mainland Asian contexts, Vietnam, Singapore, China, and multi-country samples, technological capability and digital adoption are statistically significant moderators of the I-P relationship (Goedhuys & Sleuwaegen, 2013; Teece, 2007). Why are they null in SIDS?

The FIP mechanism provides the answer: capabilities moderate *organizational complexity costs* on the descending portion of the inverted-U, but the FIP mechanism operates at a structural, pre-organizational level. A SIDS firm with ISO 9001 certification (high TCI) still faces the same 30–210% trade cost amplifications; a SIDS firm with a functioning website (DAI = 1) still faces absent trade finance institutions. Capabilities that are powerful moderators within the structural prerequisites framework become ineffective when those prerequisites do not hold.

This is consistent with the resource-based view's emphasis on context-dependency (Barney, 1991): resources generate value only within institutional environments that support appropriability. Teece's (2007) dynamic capabilities framework similarly assumes functioning market mechanisms; in their absence, dynamic capabilities may be "trapped" in the firm, unable to translate into performance advantages through international market transactions.

The positive, though non-significant, point estimates for TCI_z and DAI_z suggest that capabilities do raise general productivity levels (consistent with their role in the broader literature), but they cannot modify the negative directionality of the FSTS–performance relationship in the SIDS structural context. This nuance, capabilities help the level, not the slope, may be a productive direction for future SIDS research.

1.6.4 5.4 Policy Implications: The Antigua and Barbuda Agenda

Our findings have direct implications for SIDS development policy. The Antigua and Barbuda Agenda for SIDS (United Nations General Assembly, 2024), adopted at the Fourth International Conference on SIDS, frames export promotion and trade facilitation as central pillars of SIDS development strategy. This framing implicitly assumes that increasing FSTS will improve firm performance and thereby stimulate economic growth.

Our results caution against this assumption. If FIP is a structural feature of Pacific SIDS firm economies, then policies that push SIDS firms toward higher export intensity, without simultaneously addressing the violated structural prerequisites, may worsen firm performance at the micro-level even while improving macroeconomic trade balances. The relevant policy question becomes: what interventions can address the three structural prerequisites that generate FIP?

On **Prerequisite 1** (viable domestic market): regional integration initiatives (Pacific Agreement on Closer Economic Relations Plus, PACER Plus) that create effective market access to neighboring SIDS and to Australia and New Zealand could expand the effective domestic market beyond national borders, reducing structural compulsion to export to distant high-cost markets.

On **Prerequisite 2** (manageable trade costs): transport cost reduction through regional mar-

itime connectivity programs and, where feasible, air cargo subsidies for high-value, low-weight exports (agricultural products, crafts, services) can reduce the trade cost amplification that drives FIP.

On **Prerequisite 3** (functional institutions): targeted institutional development focused on trade facilitation infrastructure, standards bodies, trade finance intermediaries, export promotion agencies with genuine capacity, rather than generic governance reform may most directly address the FIP mechanism. Goedhuys and Sleuwaegen's (2013) evidence on international certification is instructive: their finding that ISO certification improves firm productivity is consistent with the null TCI result here if certification's productivity benefit is conditional on institutions that support standards enforcement and customer recognition, conditions absent in the SIDS context.

1.6.5 5.5 Limitations

Several limitations bound the interpretation of our findings.

First, **cross-sectional pooled structure**: while we pool across multiple country-year waves (2009–2025), few SIDS economies have more than one WBES wave (see Table 1), limiting panel identification of causal effects. The FIP estimate is primarily identified from cross-sectional variation in FSTS within country-year cells. Future work using panel data, where available for Fiji and Papua New Guinea, which have two waves, could strengthen causal identification.

Second, **FSTS measurement in thin-export contexts**: with mean FSTS of 0.060 and only 12.7% exporters, the FSTS distribution is highly right-skewed. Coefficient estimates in M2 may be sensitive to a handful of high-FSTS outliers; the non-significant quadratic results should be interpreted cautiously. Future work could employ quantile regression or winsorization to assess distributional sensitivity.

Third, **DAI as Tier-1 proxy only**: our digital adoption measure captures only website presence, a foundational, static indicator. This is appropriate for the SIDS context given the limited availability of more advanced digital infrastructure indicators in WBES, but it means we cannot assess whether more sophisticated digital capabilities (e-commerce, mobile payments, cloud computing) might moderate FIP differently. We explicitly do not claim to measure dynamic digital capability.

Fourth, **generalizability beyond Pacific SIDS**: FIP is theorized as arising when three structural prerequisites are simultaneously absent. Other extreme contexts, Saharan micro-states, remote highland economies, post-conflict settings, may exhibit similar patterns. Caribbean SIDS (not included in our sample) represent a natural extension context.

1.7 6. Conclusions

We introduced the Forced Internationalization Penalty (FIP), the monotone negative effect of internationalization intensity on firm performance when three structural prerequisites of the inverted-U relationship are simultaneously absent: a viable domestic market, manageable trade costs, and functional institutional support. Using WBES data from 1,469 firms across nine Pacific SIDS economies, we find robust support for FIP across four empirical specifications: $\beta = -0.404$, $p = .032$ in the primary country-and-year fixed-effects model; $\beta = -1.236$, $p < .001$ without country fixed effects; and $\beta = -0.901$, $p = .027$ in the exporters-only subsample. The quadratic term is not significant, confirming the monotone negative form with no observable turning point. Technological capability and digital adoption, significant in mainland Asian contexts, are null in the SIDS environment, consistent with the FIP theoretical mechanism.

These findings contribute to IB theory in two ways. First, they help reorient the discourse on institutional effects from "institutional moderation of I-P intensity" to "institutional determination of I-P sign," suggesting that the inverted-U may be conditional on structural prerequisites rather than broadly applicable across contexts. Second, they extend the application of "forced internationalization" beyond the SME literature into the performance-outcome domain, providing a theoretically defined and empirically grounded construct (FIP) that future research can build upon.

For the world development policy agenda, FIP suggests that export promotion strategies in SIDS require complementary structural interventions addressing trade costs, domestic market viability through regional integration, and targeted institutional development, not merely the intensification of export activity per se.

1.8 References

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1.9 Appendix A: Tables

1.9.1 Table 1. Country-Level Summary Statistics

Economy	N (firms)	Exporters	Exporter %	Mean FSTS	Waves
Fiji	—	—	—	—	—
Kiribati	—	—	—	—	—
Papua New Guinea	—	—	—	—	—
Samoa	—	—	—	—	—
Solomon Islands	—	—	—	—	—
Timor-Leste	—	—	—	—	—
Tonga	—	—	—	—	—
Vanuatu	—	—	—	—	—
Comoros	—	—	—	—	—

Economy	N (firms)	Exporters	Exporter %	Mean FSTS	Waves
Total	1,469	187	12.7%	0.060	2009–2025

Note: Country-level breakdowns are available in the replication archive (p8/replication/). Total N reflects analysis sample (non-missing ln_LP and FSTS). Wave coverage varies by country.

1.9.2 Table 2. Regression Results: Internationalization Intensity and Labor Productivity

	M0 Controls	M1 Linear	M2 Quadratic	M3 Capabilities	M_yearFE
FSTS_c		−0.404* (0.188)	−0.925 (0.828)	−0.974 (0.850)	
FSTS_c²			+0.649 (0.980)	+0.782 (1.001)	
FSTS					−1.236*** (0.269)
TCI_z				+0.058 (0.085)	
DAI_z				+0.062 (0.073)	
ln_size	+0.080 (0.070)	+0.082 (0.070)	+0.084 (0.070)	+0.062 (0.070)	+0.247* (0.099)
firm_age	+0.009* (0.005)	+0.009. (0.005)	+0.009. (0.005)	+0.010* (0.005)	+0.047*** (0.014)
foreign_own_pct	−0.006. (0.003)	−0.006. (0.003)	−0.006. (0.003)	−0.006. (0.003)	−0.006* (0.003)
Country FE	✓	✓	✓	✓	—
Year FE	✓	✓	✓	✓	✓
N	1,469	1,469	1,469	526	1,469
R ²	0.799	0.800	0.800	—	0.512

*Note: Coefficients reported with heteroscedasticity-robust standard errors in parentheses. Country fixed effects absorbed in M0–M3. *** $p < .001$; ** $p < .01$; * $p < .05$; . $p < .10$.*

Forced Internationalization Penalty (FIP): Confirmed, $\beta(\text{FSTS_c}) = -0.404$, $p = .032$ (M1).

1.9.3 Table 3. Robustness: Exporters-Only Subsample (N = 187)

Specification	$\beta(\text{FSTS_c})$	SE	p-value	Direction
OLS, country+year FE, N=187	−0.901	0.398	.027*	Negative
M1 full sample	−0.404	0.188	.032*	Negative
M_yearFE full sample	−1.236	0.269	<.001***	Negative
M_bivariate	−1.596	0.263	<.001***	Negative

Note: Penalty confirmed in all specifications. Exporters-only result ($\beta = -0.901$) addresses adverse selection concern: among active exporters, intensifying exports further worsens performance.